

Jacob Keller

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Education

Olin College of Engineering – Needham, MA

May 2028

- BS, Electrical and Computer Engineering, GPA: 3.91
- Relevant Coursework: Embedded Software (**C**), Software Systems (**C**), Software Design (**Python**), PCB Design (**KiCAD**), Computer Architecture (**System Verilog**), Modeling and Simulation (**MATLAB**), Linear Algebra, Discrete Mathematics

Experience

Rivian and Volkswagen Group Technologies – Palo Alto, CA

Embedded Software Engineering Intern

May 2026 – Aug 2026

- Developed firmware in MISRA-compliant **C** (and **Bazel**) to measure timing of the vehicle's sleep-to-wake sequence
- Merged to main and live in production on every Rivian R2, capturing 16,000+ events weekly across 460+ unique VINs
- Compiled fleet-wide results in custom dashboard, automating the previous per-vehicle measurement workflow
- Ported sleep-wake orchestration firmware (**C**, **Bazel**) to upcoming VW Group vehicle platforms, delivered ahead of an in-person technical demo for VW Group engineers visiting from Germany
- Validated software by developing **gtest** firmware unit tests, creating **pytest** HIL test cases, and bench/vehicle testing

Olin Formula SAE Electric – Needham, MA

Electrical & Software Lead

May 2026 – Present

- Own system-level design of electrical and software architecture for a high performance FSAE electric racecar
- Lead 35+ engineers to design, build, test, and race the fastest electric racecar in school history

Low Voltage Lead

May 2025 – May 2026

- Led 5 engineers across 9 projects spanning **C** firmware for STM32, PCB design (**KiCAD**), and **Python** telemetry
- Re-factored legacy CAN API (**C**, **Bazel**) to transition from 8-bit AVR to 32-bit ARM architecture (STM32). Replaces repetitive and error-prone individual CAN implementations with a unified collection of YAML description files

Integration Engineer

Sept 2024 – May 2025

- Developed embedded **C** firmware for ATmega16M1 microcontroller. Utilized CAN, SPI, and PWM protocols to measure and visually communicate low voltage system status and control the pump/fan within the cooling loop
- Created **Python** telemetry system using UART radios, enabling wireless diagnostics for the first time in team history

Formosa EV – Orange, CA

EV Engineering Intern

Mar 2025 – Aug 2025

- First intern and early employee at startup developing modern EV conversion components
- Converted 1965 Corvair to electric by adapting original throttle linkage to an electronic sensor using custom designed **3D-printed** parts. Designed, welded, and integrated custom sheet metal mounting for motor, batteries, and radiator

Projects

Bare-Metal Bicycle Safety Light, jacobkeller.me/bike

Sep 2026

- Developed bare-metal embedded **C** firmware for STM32 to sequence through a finite state machine of lighting modes
- Configured hardware and timer interrupts via memory-mapped registers for interrupt-driven state transitions

Portable Diagnostic PCB, jacobkeller.me/diag

Jun 2026

- Designed a new compact PCB (**KiCAD**) to quickly view the status of a Formula SAE electric racecar without requiring a laptop. Represents the status of each node in the vehicle's shutdown circuit with an easy-to-understand LED
- Optimized design to protect integrity of CAN and SPI signals and minimize disruptions to the ground plane

Skills and Interests

Programming: C, Python, System Verilog, MATLAB

Tools: Git, Linux, pytest, gtest, Bazel, CMake, KiCAD, Databricks, Onshape, Solidworks, Oscilloscopes, Logic Analyzers

Interests: Backpacking, Rock Climbing, EV Conversions, Open-Source Hardware, Walkable Cities, History